EXERCICE 1

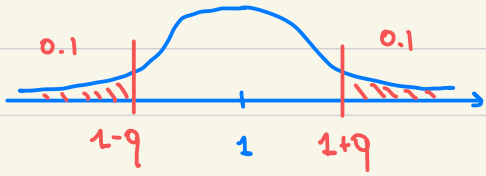
ⓐ $P(|X|<2) = P(X<2) - P(X<-2) = \text{pnorm}(2,1,2) - \text{pnorm}(-2,1,2)$

$= 0.6247$

ⓑ $P(X^2>10) = P(X<-\sqrt{10}) + P(X>\sqrt{10}) = \text{pnorm}(-\sqrt{10},3,\sqrt{0.4}) + 1$

$- \text{pnorm}(\sqrt{10},3,\sqrt{0.4}) = 0.3988$

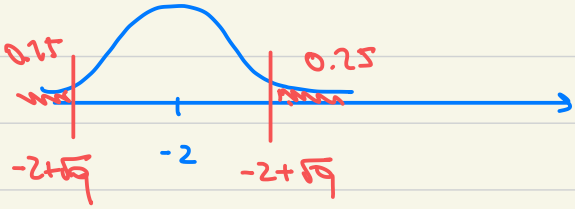
ⓒ $P(|X-1|\geq q) = 0.2$



$1+q = \text{qnorm}(0.9, 1, \sqrt{3})$

$\Rightarrow q = 2.2197$

ⓓ $P((X+2)^2 \geq q) = 0.5 \iff P(X+2 \geq \sqrt{q})$



$-2+\sqrt{q} = \text{qnorm}(0.75, -2, 3)$

$\Rightarrow q = 4.0944$

ⓔ $P(|X|<q) = 0.80$

$P(|X|<q) = P(X<q) - \underbrace{P(X<-q)}_{\approx 0} = 0.80$

$\Rightarrow q = \text{qnorm}(0.80, 20, \sqrt{2}) =$

EXERCISE 2

ⓐ $P(X<3) = \text{pexp}(3,1) = 0.9502$

ⓑ $P(1 \le X^2 \le 3) = P(1 < X < \sqrt{3}) = \text{pexp}(\sqrt{3}, 0.5) - \text{pexp}(1, 0.5) = 0.1859$

ⓒ $P(|X|>4) = P(X>4) = 1 - \text{pexp}(4, 0.2) = 0.4493$

ⓓ $P(X \ge q) = 0.05 \qquad q = \text{qexp}(0.95, 0.2) = 14.9787$

ⓔ $P(\exp(X+1) \le q) = 0.85$

$\iff P(X \le \ln q - 1) = 0.85$

$\ln q - 1 = \text{qexp}(0.85, 2) = 0.94856$

$q = \exp((0.94856+1)) = 7.0185$

EXERCISE 3

ⓐ $P(X<3) = \int_1^3 \underbrace{\frac{1}{4-1}}_{\text{pdf of } U(1,4)} dx = \frac{3-1}{3} = 2/3$

otherwise $= \text{punif}(3,1,4) = 0.66$

ⓑ $P(1 < X^2 < 3) = P(-\sqrt{3} < X < -1 \text{ or } 1 < X < \sqrt{3})$

$= \int_{-\sqrt{3}}^{-1} \frac{1}{8} dx + \int_1^{\sqrt{3}} \frac{1}{8} dx = \frac{2(\sqrt{3}-1)}{8} = 0.1830$

otherwise $= \text{punif}(\sqrt{3}, -2, 6) - \text{punif}(1, -2, 6) + \text{punif}(-1, -2, 6) - \text{punif}(-\sqrt{3}, -2, 6) = 0.1830$

ⓒ $P(X^2 > 4) = P(X<-2) + P(X>2) = \int_{-4}^{-2} \frac{1}{6+4} + \int_2^6 \frac{1}{6+4} dx = \frac{6}{10}$

also $= \text{punif}(-2, -4, 6) + 1 - \text{punif}(2, -4, 6) = 0.6$

EXERCISE 4

ⓐ $P(X < 0.85) =$ pgamma(0.85, 1, 2) $= 0.8173165$

ⓑ $P(|X| > 4) =$ 1 − pgamma(4, 5, 1) $= 0.6288$

ⓒ $P(X \geq q) = 0.05$ $\quad q =$ qgamma(0.95, 5, 1) $= 9.1535$

ⓓ $P(\sqrt{X+1} \geq q) = 0.85$

$P(X \geq q^2 - 1) = 0.85$

$q^2 - 1 =$ qgamma(0.15, 2, 3) $= 0.2278$

$q = \sqrt{0.2278 + 1} = 1.1080$

ⓔ sim = rgamma(10000, 1, 3)

mean(sim)